Looped Transformers: improving reasoning through iterative computation

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Abstract

Looped Transformers increase inference-time computation by repeatedly applying shared Transformer layers without increasing the number of model parameters. This project investigates whether additional iterations improve language model reasoning and whether the improvements justify their computational cost. We compare two models trained for recurrent computation, Ouro-2.6B and Huginn-3.5B, with Qwen3-4B, a standard pretrained Transformer modified to support training-free looping. We evaluate different loop counts and layer selections on mathematical and science reasoning benchmarks: GSM8K, MATH-500, and ARC-Challenge. Performance will be assessed using answer accuracy, inference latency, and computational cost. The project aims to provide a reproducible implementation of inference-time looping and analyze its effectiveness across models with comparable parameter counts.

1 Scientific Goal of the Project

Problem Description. Standard Transformers process inputs through a fixed sequence of layers. Improving reasoning performance often involves using larger models or generating additional reasoning tokens, both of which increase computational requirements. Looped Transformers offer an alternative by repeatedly applying shared layers, increasing computation without adding parameters (Geiping et al., 2025; Zhu et al., 2026; Azizi and others, 2026).

However, models trained for recurrent computation may behave differently from standard pretrained models whose layers are repeated without retraining. It remains important to establish when additional iterations improve reasoning and whether the benefits outweigh the inference costs.

Our goal is to systematically compare native and training-free looping across models with similar parameter budgets. We will investigate the effect of loop count and layer selection on reasoning accuracy, while measuring inference computation and runtime. The resulting comparisons will help identify the conditions under which iterative computation is beneficial.

Research Questions. This project investigates whether iterative computation improves language model reasoning without increasing the parameter budget. We address three research questions: **(RQ1)** Does increasing recurrent depth improve reasoning accuracy in models trained for looping, such as Ouro and Huginn? **(RQ2)** Can training-free looping improve reasoning performance in Qwen3-4B, and how does the selection of repeated layers affect the results? **(RQ3)** How do looped

and conventional Transformers with comparable parameter counts differ in reasoning accuracy and inference cost, and do the improvements justify the additional computation?

2 Significance of the Project

Motivation and Contribution. Improving language model reasoning often requires additional computational resources. Looped Transformers offer an alternative to increasing model size by allowing repeated computation using shared parameters (Geiping et al., 2025; Zhu et al., 2026). However, the benefits of additional iterations may depend on the architecture and inference cost. Our contribution will be a reproducible comparison of native and training-free looping across models with similar parameter counts. By evaluating reasoning accuracy alongside computational cost, we aim to identify when iterative computation is beneficial and when its additional cost outweighs the gains.

3 Evaluation Datasets

We propose three publicly available benchmarks: GSM8K (Cobbe et al., 2021) for multi-step arithmetic reasoning, MATH-500 (Hendrycks et al., 2021; Lightman et al., 2024) for advanced mathematical reasoning, and ARC-Challenge (Clark et al., 2018) for multiple-choice science reasoning. The datasets are summarized in Table 1.

Dataset	Test / Total	Licence
GSM8K	1,319 / 8,792	MIT
MATH-500	500 / 500	MIT*
ARC-Challenge	1,172 / 2,590	CC BY-SA 4.0

Table 1: Proposed evaluation benchmarks. *MIT applies to the upstream MATH repository; rights to some original problems are uncertain.

GSM8K consists of 7,473 training and 1,319 test examples, containing grade-school mathematics problems with step-by-step solutions and numerical answers. **MATH-500** contains 500 evaluation problems with numerical or symbolic answers, categorized by mathematical subject and difficulty level. **ARC-Challenge** consists of 1,119 training, 299 validation, and 1,172 test questions, each with multiple answer choices and a correct answer identifier.

Conventional class balance is not applicable to GSM8K or MATH-500, as their answers are open-ended. In ARC-Challenge, answer labels identify choices rather than fixed semantic classes. Their distribution will be examined in Phase 2.

The datasets are available from Hugging Face: `openai/gsm8k`, `HuggingFaceH4/MATH-500`, and `allenai/ai2_arc`. All three are classified as **READY** for access. Evaluation will use only the designated test splits; no model training or fine-tuning is planned.

4 Paper summaries

Table 2 synthesizes 16 key studies across training-free looping, native recurrent pre-training, parameter sharing, and latent reasoning, detailing each method's architectural approach, input representation, evaluation datasets with explicit splits, and reported numerical outcomes. Across these works, recurrent depth consistently improves multi-step reasoning and algorithmic length generalization while decoupling parameter count from computational depth. Crucially, the literature demonstrates that while naive test-time looping in frozen models destabilizes representations, controlled numerical integration, state interpolation, or learned halting reliably unlock latent computation without retraining from scratch.

5 Experimental pipeline figure

Figure 1 illustrates the end-to-end experimental pipeline, connecting input benchmark test splits, the training-free mid-stack looping wrapper, inference-time halting control, and the evaluation methodology for reasoning accuracy and inference cost.

6 Model Selection

We select three open-weight language models with comparable parameter counts: Qwen3-4B (Qwen Team, 2025), Ouro-2.6B (Zhu et al., 2026), and Huginn-3.5B (Geiping et al., 2025). Qwen3-4B serves as the standard Transformer baseline and as the model for training-free looping. Ouro and Huginn were trained with recurrent computation, allowing us to evaluate models designed for iterative processing. Their architectures, parameter counts, and estimated memory requirements are summarized in Table 3.

All three model checkpoints are publicly available on Hugging Face under the Apache 2.0 licence. We have access to a GPU with 12 GB of

Table 2: Prior evidence on looped, recurrent-depth, and parameter-shared Transformers. Values are the authors' reported metrics on the specified dataset and split; different tasks, inputs, and evaluation conventions prevent direct ranking. CoT = chain-of-thought; dev = development; PPL = perplexity.

Work	Approach and input	Dataset and split	Reported outcome
Chen et al. 2026 (Chen et al., 2026)	Training-free mid-stack ($W = 4$) K-stage RK sub-stepping ($h = 1/K$, β-anchor) + 2-phase KV cache	MMLU-Pro 5-shot CoT test, GPQA-Main 0-shot test, ARC-C 25-shot test	$+2.64$ pp on MMLU-Pro ($57.14\% \rightarrow 59.79\%$), $+2.01$ pp on GPQA-Main with `Qwen3-4B-Instruct`; $+2.30$ pp on ARC-C
Lys et al. 2026 (Lys et al., 2026)	Training-free middle-looping ($R = 3$, 40–60% depth) with Uniform / EMA / Auto-align state interpolation	GSM8K 5-shot test, WinoGrande 5-shot test, ARC-C 25-shot test	`Gemma-2-2B` gains $+1.36$ pp on GSM8K ($25.02\% \rightarrow 26.38\%$), $+0.78$ pp on WinoGrande, $+0.59$ pp on ARC-C
Geiping et al. 2025 (Geiping et al., 2025)	`Huginn-0125` (3.5B) recurrent depth $(2, 4, 2)$ with concat adapter, sandwich norm, loop scaling $r = 1..32$	ARC-E 0-shot test, ARC-C 25-shot test, GSM8K 8-shot CoT test	ARC-E 69.91% (vs 34.89% at $r = 1$), ARC-C 38.23% (vs 24.06%), GSM8K CoT 42.08% (vs 0.00% at $r = 1$ and 2.20% twin)
Zhu et al. 2026 (Zhu et al., 2026)	`Ouro` (1.4B/2.6B) whole-stack LoopLM ($T_{\text{max}} = 4$) with uniform KL entropy loss and loss-gain Q-Exit gate	MATH500 5-shot CoT test, BBH 3-shot CoT test, MMLU-Pro 5-shot CoT test	`Ouro-2.6B` ($T = 4$) scores 90.85% on MATH500 (vs `Qwen3-8B` 62.30%), 80.46% on BBH, 55.73% on MMLU-Pro
Bae et al. 2025 (Bae et al., 2025)	Relaxed Recursive Transformers; layer tying across depth blocks with SVD-initialized per-loop LoRA + depth batching	Few-shot test suite (HellaSwag, PIQA, WinoGrande, ARC-E/C), 60B-token uptraining split	`Recursive Gemma-1B` gains $+13.5$ pp accuracy (22% error reduction) over from-scratch 1B baseline; 2–3$\times$ throughput speedup
Dehghani et al. 2019 (Dehghani et al., 2019)	Universal Transformer; shared multi-head self-attention block iterated across depth with per-token ACT halting	WMT14 En-De newstest2014 test set, bAbI 10k QA test set	28.9 BLEU on WMT14 En-De (vs 28.0 vanilla Transformer); 0.23% mean error (0 failed tasks) on bAbI
Lan et al. 2020 (Lan et al., 2020)	ALBERT; cross-layer parameter sharing across all blocks with factorized embeddings and SOP loss	GLUE benchmark test set, SQuAD v2.0 test set, RACE test set	`ALBERT-xxlarge` scores 89.4 on GLUE, 92.2 F1 on SQuAD v2.0, 86.5% on RACE with 70% fewer parameters than BERT-large
Csordás et al. 2024 (Csordás et al., 2024)	MoEUT; weight-shared Universal Transformer layers with sparse MoE routing and peri-layernorm	C4 held-out validation set, LAMBADA test split, HellaSwag validation split	Perplexity 13.24 on C4 at 244M params (vs 13.40 baseline); 30.6% on LAMBADA; 65.2% on HellaSwag with 50–80% MACs
Saunshi et al. 2025 (Saunshi et al., 2025)	Looped Transformer with latent thoughts; identical weight-tied layers applied iteratively for multi-step reasoning	i-GSM synthetic math reasoning test set, modular arithmetic (p-hop) induction test split	Depth-8 model looped $L = 2$ times achieves 73.2% accuracy on i-GSM (matching depth-16 non-looped model 72.8%)
Hao et al. 2024 (Hao et al., 2025)	Coconut; continuous latent thought tokens fed back into context without discrete language projection	GSM8K test set, ProntoQA 5-hop test set, ProsQA test split	34.1% accuracy on GSM8K (outperforming No-CoT 16.5% and matching explicit CoT); higher reasoning accuracy with fewer tokens
Men et al. 2025 (Men et al., 2025)	ShortGPT; pruning redundant middle layers based on Block Influence (BI) metrics without fine-tuning	MMLU 5-shot test set, GSM8K 8-shot test set (`LLaMA-2-70B`, `Mistral-7B`)	Retains over 95% baseline accuracy on MMLU (67.4% vs 68.9%) and GSM8K (53.8% vs 54.5%) when pruning 25% of mid-layers
Lad et al. 2024 (Lad et al., 2024)	Layer ablation and swapping across four inference stages (detokenization, feature engineering, ensembling, sharpening)	OpenBookQA test set, ARC-Easy test set, `LLaMA-2-7B` / `Pythia` splits	Layer 0 ablation drops accuracy to chance ($< 25\%$), whereas swapping adjacent mid-layers (layers 12–20) causes $< 1.5\%$ degradation
Balagansky & Gavrilov 2022 (Balagansky and Gavrilov, 2022)	PALBERT; introduces adaptive computation time and Q-Exit thresholding to ALBERT's parameter-shared layers	GLUE benchmark development sets (MNLI, QQP, QNLI, SST-2)	Matches baseline ALBERT score (82.8 vs 82.3 average score) while executing 35% fewer layer iterations on average
Fan et al. 2025 (Fan et al., 2025)	Looped Transformers with input-dependent adaptive halting for length generalization on algorithmic problems	Modular arithmetic, parity, and sequence copy tasks; trained on length 20, tested on length 100+	100% test accuracy on length 100 sequences where standard fixed-depth Transformers collapse to 0%
Bansal et al. 2022 (Bansal et al., 2022)	Recurrent deep thinking networks with continuous input injection and randomized unrolling during training	Prefix sums and 2D maze navigation; hard test splits (2$\times$–5$\times$ longer than training)	Achieves 99.8% test accuracy on extrapolated problem lengths with persistent input injection (vs 2.1% without)
Giannou et al. 2023 (Giannou et al., 2023)	Looped Transformer configured as a programmable computer by storing instruction pointers and memory in residual stream	Synthetic programmatic execution traces (matrix multiplication, bubble sort); exact match split	100% execution accuracy on bounded programmatic subroutines in logarithmic looped steps

VRAM, which may support inference with limited batch sizes and sequence lengths. However, actual memory requirements may be higher, particularly if higher numerical precision is used. If necessary, we may use external GPU providers.

7 Evaluation Methodology

Reasoning Accuracy. All models will be evaluated on the same benchmark test splits using generation-based evaluation. Final answers will be extracted from model outputs and compared with reference answers. GSM8K will use

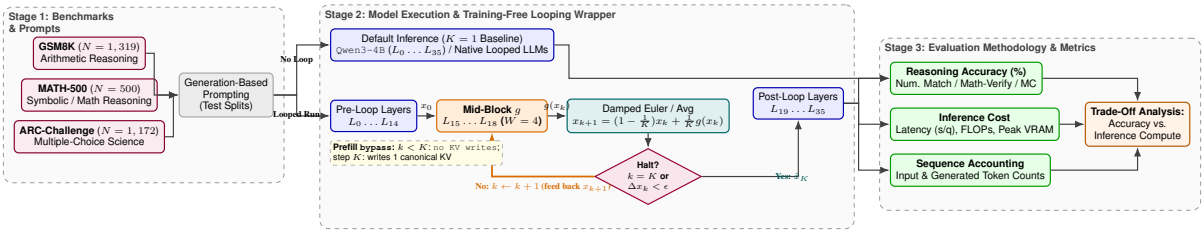


Figure 1: End-to-end experimental pipeline: Stage 1 prepares test-split inputs across GSM8K, MATH-500, and ARC-Challenge for generation-based evaluation; Stage 2 executes frozen `Qwen3-4B` (alongside native looped baselines `Ouro-2.6B` and `Huginn-3.5B`) across varying recurrence depths and layer configurations; Stage 3 extracts final answers (via normalized numerical matching, Math-Verify, and multiple-choice matching) and records latency, FLOPs, peak GPU memory, and token counts to analyze accuracy–compute trade-offs.

Model	Params	Type	BF16 weights
Qwen3-4B	4B	Standard	8.0 GB
Ouro-2.6B	2.6B	Looped	5.2 GB
Huginn-3.5B	3.5B	Recurrent	7.0 GB

Table 3: Selected models and approximate GPU memory for weights in 16-bit precision, excluding inference overhead.

normalized numerical matching, MATH-500 will use mathematical equivalence checking (Math-Verify), and ARC-Challenge will use multiple-choice accuracy. Missing or unparseable answers will count as incorrect. Accuracy, defined as the proportion of correctly answered questions, will be reported separately for each benchmark.

Inference Cost. We will measure inference latency (seconds per question), estimate floating-point operations (FLOPs), and record peak GPU memory usage. Input and generated token counts will also be recorded to account for differences in sequence length and generated response length.

Experimental Comparisons. We will first compare each model across different recurrence depths against its default inference configuration. For Qwen3-4B, we will additionally evaluate the effect of repeating different Transformer layers. Prompting, decoding settings, and output limits will remain fixed within each model's experiments.

We will then compare Qwen3-4B, Ouro-2.6B, and Huginn-3.5B across their shared benchmarks, reporting parameter counts alongside accuracy and inference cost. Since the models differ in architecture and training, cross-model comparisons will be interpreted cautiously.

We will analyze accuracy–compute trade-offs across looping configurations to determine whether additional inference computation justifies improvements in reasoning performance.

8 Work plan and deliverables

Each workstream has one *accountable owner* and one *supporting implementer/reviewer*; the supporting member contributes code, testing or interpretation[cite: 265]. All three members review five papers each and jointly prepare the report, video, responses to feedback and Q&A[cite: 265]. Work division and deliverables are presented in Table 4, and the week-by-week schedule is given in Table 5[cite: 265].

Workstream	**Owner**	**Deliverable**
Theory & Error Analysis	Rafał	SOTA taxonomy & 20-error failure-mode audit report
Evaluation & Benchmarks	Antoni	Standardized `lm-eval` suite, CI & Pareto figures
Looping Wrapper & Models	Kuba	PyTorch mid-block loop wrapper & PoC/final MRs

Table 4: Workstream division, accountable owners, and project deliverables.

Data Statement

The datasets are publicly available on Hugging Face, with sources and licences listed in Table 1. They contain educational questions and answers, with no personal data expected. We will use existing annotations without collecting new data or performing manual annotation. The pretrained models (Qwen3-4B, Ouro-2.6B, and Huginn-3.5B) are also obtained from Hugging Face under Apache

Dates	Tasks and Milestones	Team
30 Oct–4 Nov	Verify literature, licenses, and data splits; Phase-1 proposal (4 Nov)	All; Kuba edits
5–11 Nov	Loaders (`Qwen3`, `Ouro`, `Huginn`), `lm-eval` harness; $K = 1$ baselines	Antoni, Rafał, Kuba
12–18 Nov	Mid-block wrapper PoC (`Qwen3-4B` [15, 18]); feedback (13 Nov); 10-min video (18 Nov)	Kuba, Rafał
19–25 Nov	Initial runs ($K \in \{2, 4\}$) on ARC-C & GSM8K; peak VRAM check (12 GB); Q&A (25 Nov)	Rafał, Antoni
26 Nov–2 Dec	PoC merge request, Damped Euler vs. naive loop, revised report, rebuttal (2 Dec)	All; Kuba edits
3–9 Dec	Apply Phase-1 feedback; freeze layer window [15, 18] and `Math-Verify` splits	Antoni, Rafał
10–16 Dec	Main $K \in \{1, 2, 4, 8\}$ evaluation across 3 models on GSM8K, MATH-500, ARC-C; log 20 errors	Kuba, Antoni
17–23 Dec	Early-exit & Q-Exit check; accuracy vs. latency/FLOPs Pareto figures; draft discussion	Rafał, Antoni
24–30 Dec	Break / contingency buffer; no dependency on new experiments	All if needed
31 Dec–6 Jan	Cross-reproduce best configurations; review final report draft	All; Antoni verifies
7–13 Jan	Final report PDF, code repository/MR cleanup, presentation video, and final Q&A (13 Jan)	All

Table 5: Week-by-week project plan and milestone schedule.

2.0 licences (each one). Neither datasets nor model weights will be redistributed, and their licensing and attribution requirements will be respected.

Contribution Table

Member	Phase 1 Tasks Completed	Hours
Rafał Pyzwoski	Problem formulation, theoretical taxonomy, 5 theory SOTA summaries.	4 h
Antoni Adaszkiewicz	Benchmark selection, evaluation protocol, metrics, 5 SOTA summaries.	6 h
Kuba Drażan	Looping wrapper design, 5 architecture SOTA summaries, model audit, pipeline figure, schedule.	5 h

Declaration of LLM Use

Large Language Models (Google Gemini) were used to assist in formatting LaTeX code, structuring tables, and verifying bibliographic entries. All scientific designs, mathematical formulations, and experimental claims were verified against primary sources by the authors.

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